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R.J. Portacion

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N. Chelmsford, MA

Education

University of Massachusetts Lowell GPA 3.4
B.S. Computer Engineering – Shah Academy Scholar – Founder & President, IGNITE *Sep 2024 – May 2028*

Skills

Semiconductor Processing: Electroplating (ECD), LPCVD, Advanced Packaging, Wet Processing, Cleanroom
Metrology & Analysis: SEM, XRD, Profilometry, AOI, AFM, X-Ray Imaging, Defect Analysis, Hall Effect
Electronics & Hardware: PCB Design, Oscilloscopes, Equipment Troubleshooting, Multimeters, Soldering
Software & Data: JMP, Excel, C, C++, ARM Assembly, LTspice, Multisim, Quartus, MATLAB, Arduino IDE

Experience

Incoming Nanofabrication Engineer Lowell, MA
Core Research Facilities (Internship) *January 2027 – May 2027*

Process Engineer Billerica, MA
Applied Materials (Co-op) *July 2026 – December 2026*

- Optimized bottom-up Cu electroplating of 20 × 200 µm high-aspect-ratio features through DOE-driven variation of bath chemistry, temperature, dwell time, vacuum prewet, and current-density waveforms, cutting customer wafer fill time 50% from a 10-hour benchmark to 5 hours while developing wafer-level ECD and metallization processes for Cu, In, and Pd across TSV, TGV, damascene, and advanced-packaging structures to improve throughput and efficiency.
- Performed process and hardware qualification using optical profilometry, AOI, X-ray imaging, cross-sectional analysis, and defect analysis to characterize panel-level plating performance, leveraging JMP and Excel for data analysis of Cu thickness, roughness, uniformity, terminal effects, feature fill, and overburden across varying process conditions and hardware configurations to support process optimization and qualification decisions.
- Operated and troubleshot automated semiconductor ECD and wet-process equipment to sustain repeatable plating performance and equipment uptime, maintaining process cells, chemical reservoirs, vacuum systems, fluid-delivery and wafer-handling systems, electrical interfaces, and CDA/N_2 utilities while diagnosing vacuum, fluid, contact, and process-related faults through systematic equipment troubleshooting and root-cause analysis.

Research Associate (Ultra-wide Bandgap Semiconductors) Lowell, MA
Bhuiyan Research Group — Principal Investigator: Dr. Anhar Bhuiyan *Sep 2025 – July 2026*

- Supported LPCVD growth of Ga_2O_3 thin films through substrate preparation, wafer cleaving, precursor handling, reactor setup, gas-flow calibration, process monitoring, and experimental documentation to support reproducible semiconductor deposition, controlled thin-film growth, and repeatability across experimental runs.
- Characterized Ga_2O_3 thin films using XRD, SEM, AFM, and Hall Effect measurements, analyzing crystal structure, surface morphology, film quality, and electrical properties to correlate material characteristics with LPCVD growth parameters, deposition conditions, and process performance across experimental samples.
- Investigated LPCVD growth parameters and characterization results through literature review and experimental data analysis, contributing to process optimization and growth-property correlation while presenting findings and semiconductor process concepts at a university research symposium and 3-Minute Thesis competition.

Research Associate (Photonics) Lowell, MA
Wang Laboratory of Optics — Principal Investigator: Dr. Xingwei Wang *Feb 2025 – Aug 2025*

- Fabricated fiber-optic biosensors and strain sensors through optical-fiber splicing, stripping, shaping, and chemical etching using glass etchant, BOE, and HCl-based solutions to produce controlled sensor geometries.
- Developed and tested a fiber-optic glucose biosensor using optical frequency domain reflectometry (OFDR) to characterize spectral response, wavelength shifts, signal stability, and sensor performance across test conditions.

Projects

- **ARM Assembly Blackjack System – ARM Assembly, I²C, GPIO, Embedded Systems**:
  Designed an ARM-based embedded system with real-time game logic, memory management, and modular subroutines, integrating a 20×4 I²C LCD, GPIO controls, and 3.3V/5V level-shifting circuitry with hardware/software debugging.
- **Colpitts Oscillator – LTspice, Analog Circuit Design, FFT Analysis**:
  Engineered an LM741-based oscillator using LC resonance and diode feedback stabilization, iterating component values to achieve 7 kHz sinusoidal output and validating frequency response, waveform stability, and harmonic performance.